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Note

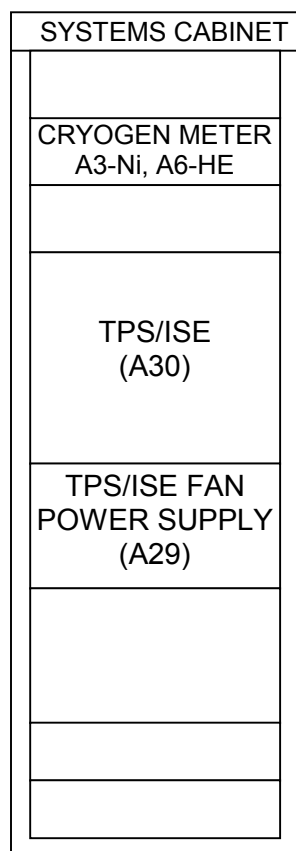
Do not start multiple copies of diagnostics. Multiple copies of diagnostics can cause different anomalies such as locked up diagnostic windows.

1- HARDWARE TESTED

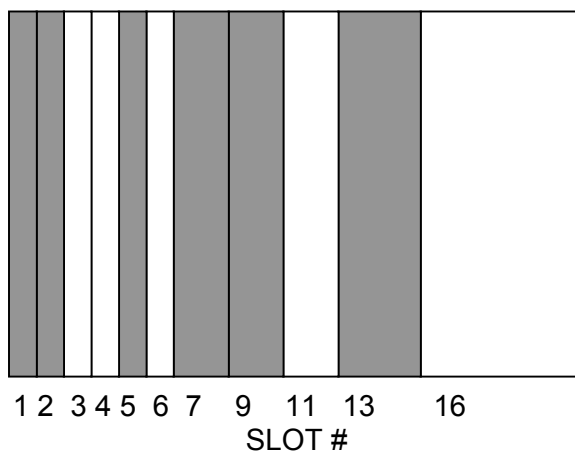
Data path tests are a "confidence" type diagnostic that check most of the TPS/ISE subsystem hardware (refer to Table 1-1 and Illustration 1-1), but in a limited way (i.e., primarily data buses). For troubleshooting, the data path test generally identifies one or more possibly faulty boards. To better isolate problems of a specific board, the TPS/ISE board level diagnostics should be run (refer to TPS/RF: Board Level Diagnostics tab).

TABLE 1-1
TPS DATA PATH HARDWARE TESTED

HARDWARE TESTS	DATA PATH
CPU Board	Yes
MEDCAM Board	Yes
VME I/F Board	Yes
CERD	Yes
IPG Board	Yes
TYME Board	Yes
Fast Rcvr	Yes



■ HARDWARE TESTED BY DATAPATH DIAGNOSTICS



- | | |
|---------------------------|-------------------------------------|
| A1 - TPS CPU | A7 - IPG or IPG-II |
| A2 - BIT3 (VME I/F) | A9 - FAST RECEIVER I/F (see Note 2) |
| A3 - BIT3 #2 (see Note 1) | A11 - SPARE |
| A4 - SPARE | A13 - CERD or UCERD |
| A5 - AP | A16 - CERD LINEAR P/S |
| A6 - SPARE | |

Note 1: Used with Signa SP workstations. Also used on pre-8.4 release Cardiac and Neuro workstations.

Note 2: Used with EchoSpeed systems.

TPS/ISE TESTED HARDWARE LOCATION
ILLUSTRATION 1-1

2- STARTING TPS / ISE DATA PATH TESTS

The Data Path tests are downloaded from the host computer to the TPS/ISE subsystem. They can be selected/run only when Signa software is running. Status/error messages are returned to the host computer for logging/display as applicable.

Data Path tests may be initiated individually or in sets. Each procedure is described below.

1. Select **[TOOLS ICON]** then **[Diagnostics]**, **[Diagnostics Main Menu]**, and **[Start...]**. Wait for the Diagnostics Main Menu to appear.
2. If desired, change iteration count: **1** (default) or Datapath Duration (time that you may select; the default is **60** seconds). The data path tests run sixty seconds for one iteration count (loop). This allows all default tests to execute at least once.

Note

A Bit 3 Data Path diagnostic button (beginning with Release 8.3) is displayed only if a Service Key is installed. Refer to procedure "Bit 3 Communication Troubleshooting". *This proprietary procedure is available for GE use, and to sites with a valid Advanced Service Package Limited License.*

3. Decide whether to run all TPS/ISE Data Path default tests, or just selected ones.

To run all default TPS/ISE Data Path Tests:

- a. Click on the Main Data Bus (see Illustration 2-1) to see the Data Path Menu (see Illustration 2-2).
- b. Select **[Select Default Datapath Tests]** from the Data Path Menu.
- c. Select **[Run Diags]** from main Diagnostics window.

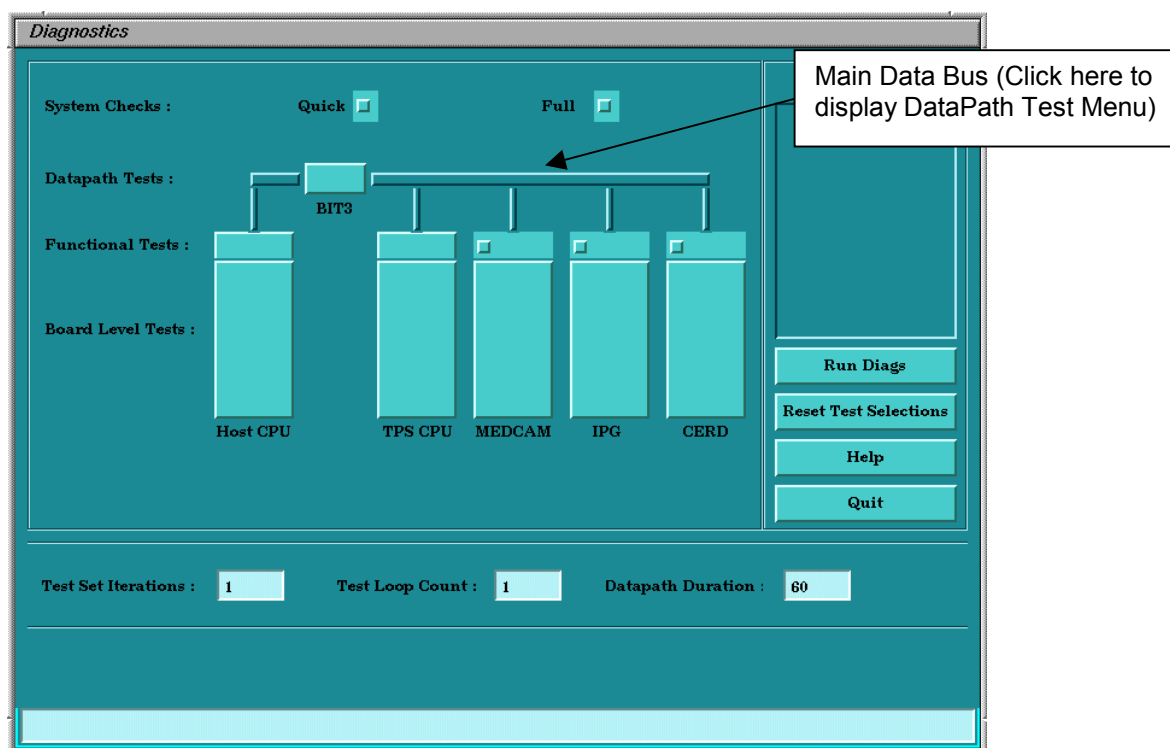
To run individual TPS/ISE Data Path Tests:

- a. Click or deselect individual Data Path tests by clicking on the box next to the appropriate test.

Note

Some test selection combinations are not allowed (i.e., they are mutually exclusive) since they use the same TPS/ISE boards. The software automatically deselects any invalid test(s) as selections are made.

- b. If desired, change the test execution time by entering a new time (in seconds) in the "Datapath Duration:" field on the main diagnostics screen. Example: 300 would run the tests for 300 seconds (five minutes).
- c. Select **[Run Diags]**. (A TPS/ISE/IPG Reset occurs before the Data Path tests start, and when your diagnostic session ends.)



DIAGNOSTICS MAIN MENU SCREEN
ILLUSTRATION 2-1



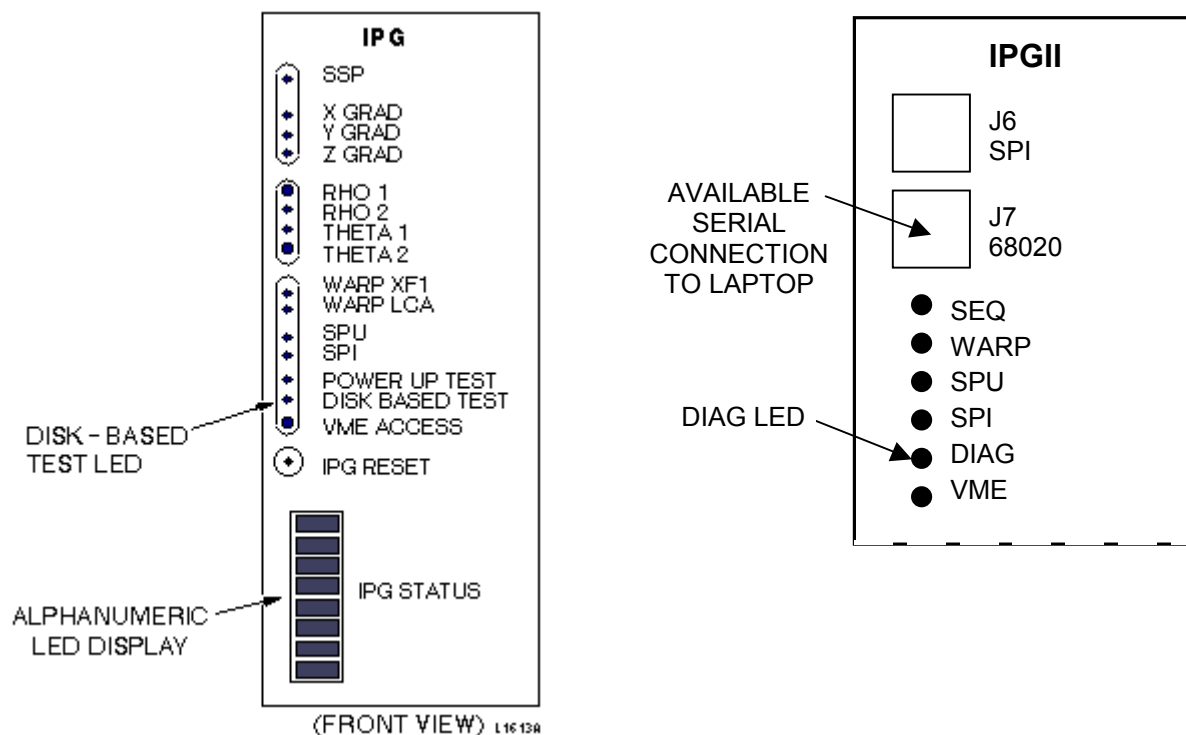
TPS/ISE DATAPATH MENU
ILLUSTRATION 2-2

3- TPS / ISE TEST INDICATORS

3-1 Local Test Indicators

IPG: Local test indicators are provided on the TPS/ISE Integrated Pulse Generator (IPG) Board. One green LED (DISK BASED TEST) and an alphanumeric display (IPG STATUS) provide the status of TPS/ISE disk-based tests and general subsystem activity. These LEDs (see Illustration 3-1) provide backup error/status indication for TPS/ISE tests in case communications with the host computer fails. While data path tests are executing, the DISK BASED TEST LED is lit, and the abbreviated name of each test is displayed on the IPG STATUS display.

IPG-II: One green LED (DIAG) and a serial connection (to laptop or PC) provides an indication of diagnostic tests (see Illustration 3-1). The LED and serial connection provide backup error/status indication for TPS/ISE tests in case communications with the host computer fails. (Refer to procedure "TPS/ISE Serial Debug Connection"; *this proprietary procedure is available for GE use, and to sites with a valid Advanced Service Package Limited License.*)



LOCATION OF IPG TEST INDICATORS
ILLUSTRATION 3-1

3-2 Operator Work Station Test Status Display

Test time remaining status is shown in the Results window:

TPS/ISE DataPath diags are
Running ...

In the DIAGS STATUS area, messages such as "TPS Reset in Progress", "Running Datapath Diags", and "Finished Diags" help indicate diagnostic test status.

3-3 Message Log

Before data path tests are run, the TPS/ISE chassis is checked to verify all expected boards (MEDCAM, CERD, and IPG) are present. If different from the last time, an error is reported to the message log. This information is stored/ updated in the TPS/ISE configuration file.

4- TPS / ISE DATA PATH TEST DESCRIPTIONS

The data path tests verify data bus communication between two or more boards via their common data buses, and exercise the various buses simultaneously to verify proper bus arbitration (i.e., really stress all the TPS/ISE communication paths). Table 4-1 lists the tests of the data paths and their associated data buses.

TABLE 4-1
TPS/ISE DATA PATHS / BUSES (TESTS UNDER TPS/ISE CPU CONTROL)

TEST NAME	DATA PATH	BUS
TPS-MEDCAM VME	TPS/ISE CPU - MEDCAM	VME
TPS-CERD VME	TPS/ISE CPU - CERD	VME
TPS-IPG VME	TPS/ISE CPU - IPG CPU	VME
IPG-TPS VME	IPG CPU - TPS/ISE CPU	VME
IPG-MEDCAM VME	MEDCAM - IPG CPU	VME
IPG-CERD SSP	CERD - IPG CPU	SSP
IPG-CERD RF	CERD - IPG CPU	RF
IPG-CERD B0 Compensation	CERD - IPG CPU	B0
MEDCAM-TPS VME	MEDCAM - TPS/ISE CPU	VME
CERD-MEDCAM VSB	CERD - MEDCAM	VSB
CERD-SPU XD Bus Monitor	CERD - IPG CPU	XD
HOST-TPS/CPU	HOST CPU - Bit3 - TPS/ISE CPU	SBus
HOST-MEDCAM	HOST CPU - Bit3 - MEDCAM	VME/SBus
HOST-IPG CPU	HOST CPU - Bit3 - IPG CPU	VME/SBus
Fast Receiver - CERD XD Bus	CERD - Fast RCVR	XD

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	May 29, 1998	J. Saperstein	Initial conversion from Toolbook to Word
1	Oct 14, 1999	M. Keber	Corrections to TPS Hardware tested illustration; added Main Diagnostics and Datapath menu illustrations, IPG-II information, and proprietary feature statement.
2	May 24, 2000	M. Jones	Added info on pre 8.4 release Cardiac and Neuro workstations to Note 1, Illustration 1. Corrected messages in DIAGS STATUS area.